

EECS 453 Principles of Machine Learning

Linear Algebra Review¹

University of Michigan, Ann Arbor

Zifei Bai

Based primarily on slides by Huijie Zhang (Hugh)

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¹Reference: CS229 Linear Algebra Review and Reference

Linear Algebra Review

- Basic Concepts and Notation
- Matrix Multiplication
- Operations and Properties
- Matrix Calculus
- Open questions if time left

Linear Algebra Tricks

- Clear about the notations
- Always check the shapes
- Use toy examples for a trail

Basic Concepts and Notation

Consider the following system of equations:

$$\begin{aligned}4x_1 - 5x_2 &= -13 \\ -2x_1 + 3x_2 &= 9.\end{aligned}$$

In matrix notation, we can write the system compactly as

$$\mathbf{Ax} = \mathbf{b}$$

with

$$\mathbf{A} = \begin{bmatrix} 4 & -5 \\ -2 & 3 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} -13 \\ 9 \end{bmatrix}$$

Exercise: Use reduced row echelon form (RREF) to solve this system.

Solution: Solving the System with RREF

Augment the coefficient matrix with $\mathbf{b}$ and apply elementary row operations:

$$\begin{aligned} \left[\begin{array}{cc|c} 4 & -5 & -13 \\ -2 & 3 & 9 \end{array} \right] &\xrightarrow{R_1 \leftarrow R_1 + 2R_2} \left[\begin{array}{cc|c} 0 & 1 & 5 \\ -2 & 3 & 9 \end{array} \right] \\ &\xrightarrow{R_2 \leftarrow R_2 - 3R_1} \left[\begin{array}{cc|c} 0 & 1 & 5 \\ -2 & 0 & -6 \end{array} \right] \\ &\xrightarrow{R_2 \leftarrow -\frac{1}{2}R_2, R_1 \leftrightarrow R_2} \left[\begin{array}{cc|c} 1 & 0 & 3 \\ 0 & 1 & 5 \end{array} \right]. \end{aligned}$$

The left side is now the identity matrix, so the last column gives the solution:

$$\boxed{x_1 = 3, \quad x_2 = 5}.$$

Basic Concepts and Notation

- By $A \in \mathbb{R}^{m \times n}$ we denote a matrix with m rows and n columns, where the entries of A are real numbers.
- By $x \in \mathbb{R}^n$, we denote a vector with n entries.
 - x column vector
 - x^T row vector
- The i th element of a vector x is denoted x_i :

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

Basic Concepts and Notation

- We use the notation a_{ij} (or A_{ij} , $A_{i,j}$, etc) to denote the entry of A in the i th row and j th column:

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

Basic Concepts and Notation

- We denote the j th column of A by a^j or $A_{:,j}$:

$$A = \left[\begin{array}{c|c|c|c} & & & \\ a^1 & a^2 & \dots & a^n \\ & & & \end{array} \right]$$

- We denote the i th row of A by a_i^T or $A_{i,:}$:

$$A = \left[\begin{array}{c} - & a_1^T & - \\ - & a_2^T & - \\ & \vdots & \\ - & a_m^T & - \end{array} \right]$$

Matrix Multiplication

The product of two matrices $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{B} \in \mathbb{R}^{n \times p}$ is the matrix

$$\mathbf{C} = \mathbf{AB} \in \mathbb{R}^{m \times p},$$

where

$$c_{ij} = \sum_{k=1}^n a_{ik} b_{kj}.$$

Note that in order for the matrix product to exist, the number of columns in $\mathbf{A}$ must equal the number of rows in $\mathbf{B}$.

Intuition

$\mathbf{AB}$ means “apply $\mathbf{B}$ first, then apply $\mathbf{A}$.” The shared inner dimension is the intermediate representation passed from $\mathbf{B}$ to $\mathbf{A}$.

Practice: Check the Shapes

Let

$$\mathbf{A} \in \mathbb{R}^{2 \times 3}, \quad \mathbf{B} \in \mathbb{R}^{3 \times 4}, \quad \mathbf{x} \in \mathbb{R}^3.$$

For each expression below, decide whether it is defined. If it is, give its shape.

- ① $\mathbf{Ax}$
- ② $\mathbf{A}^T \mathbf{x}$
- ③ $\mathbf{AB}$
- ④ $\mathbf{BA}$

Solution: Check the Shapes

- ① $A\mathbf{x}$ is defined: $(2 \times 3)(3 \times 1) = (2 \times 1)$.
- ② $A^T\mathbf{x}$ is not defined: $(3 \times 2)(3 \times 1)$ has mismatched inner dimensions.
- ③ AB is defined: $(2 \times 3)(3 \times 4) = (2 \times 4)$.
- ④ BA is not defined: $(3 \times 4)(2 \times 3)$ has mismatched inner dimensions.

Takeaway

For $(m \times n)(n \times p)$, the inner dimensions must match and the output has shape $m \times p$.

Matrix Multiplication: Vector-Vector Products

Given two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, the quantity $\mathbf{x}^T \mathbf{y}$, sometimes called the inner product or *dot* product of the vectors, is a real number given by

$$\mathbf{x}^T \mathbf{y} \in \mathbb{R} = \begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \sum_{i=1}^n x_i y_i.$$

The inner product measures alignment. It is positive when the vectors point in similar directions, zero when they are orthogonal, and negative when they point in opposing directions. Also, $\mathbf{x}^T \mathbf{y} = \mathbf{y}^T \mathbf{x}$.

Matrix Multiplication: Vector-Vector Products

Given vectors $\mathbf{x} \in \mathbb{R}^m, \mathbf{y} \in \mathbb{R}^n$ (not necessarily of the same size), $\mathbf{xy}^T \in \mathbb{R}^{m \times n}$ is their outer product. It expands two vectors into a grid of pairwise products, with $(\mathbf{xy}^T)_{ij} = x_i y_j$:

$$\mathbf{xy}^T \in \mathbb{R}^{m \times n} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{bmatrix} \begin{bmatrix} y_1 & y_2 & \cdots & y_n \end{bmatrix} = \begin{bmatrix} x_1 y_1 & x_1 y_2 & \cdots & x_1 y_n \\ x_2 y_1 & x_2 y_2 & \cdots & x_2 y_n \\ \vdots & \vdots & \ddots & \vdots \\ x_m y_1 & x_m y_2 & \cdots & x_m y_n \end{bmatrix}$$

Practice: Inner Product or Outer Product?

Let

$$\mathbf{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}.$$

- 1 Compute $\mathbf{x}^T \mathbf{y}$ and state its shape.
- 2 Compute $\mathbf{x} \mathbf{y}^T$ and state its shape.
- 3 Are the two products interchangeable?

Solution: Inner Product or Outer Product?

The inner product is a scalar:

$$\mathbf{x}^T \mathbf{y} = 1(3) + 2(-1) = 1 \in \mathbb{R}.$$

The outer product is a matrix:

$$\mathbf{xy}^T = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \begin{bmatrix} 3 & -1 \end{bmatrix} = \begin{bmatrix} 3 & -1 \\ 6 & -2 \end{bmatrix} \in \mathbb{R}^{2 \times 2}.$$

Takeaway

The position of the transpose changes both the shape and the meaning of the result.

Matrix Multiplication: Vector-Vector Products

As an example of how the outer product can be useful, let $\mathbf{1} \in \mathbb{R}^n$ denote an n -dimensional vector whose entries are all equal to 1. Furthermore, consider the matrix $A \in \mathbb{R}^{m \times n}$ whose columns are all equal to some vector $\mathbf{x} \in \mathbb{R}^m$. Using outer products, we can represent A compactly as,

$$A = \begin{bmatrix} | & | & \cdots & | \\ \mathbf{x} & \mathbf{x} & \cdots & \mathbf{x} \\ | & | & \cdots & | \end{bmatrix} = \begin{bmatrix} x_1 & x_1 & \cdots & x_1 \\ x_2 & x_2 & \cdots & x_2 \\ \vdots & \vdots & \ddots & \vdots \\ x_m & x_m & \cdots & x_m \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{bmatrix} \begin{bmatrix} 1 & 1 & \cdots & 1 \end{bmatrix} = \mathbf{x}\mathbf{1}^T$$

Matrix Multiplication: Matrix-Vector Products

Given a matrix $A \in \mathbb{R}^{m \times n}$ and a vector $\mathbf{x} \in \mathbb{R}^n$, their product is a vector $\mathbf{y} = A\mathbf{x} \in \mathbb{R}^m$.

- If we write A by rows, then we can express $A\mathbf{x}$ as,

$$\mathbf{y} = A\mathbf{x} = \begin{bmatrix} - & \mathbf{a}_1^T & - \\ - & \mathbf{a}_2^T & - \\ & \vdots & \\ - & \mathbf{a}_m^T & - \end{bmatrix} \mathbf{x} = \begin{bmatrix} \mathbf{a}_1^T \mathbf{x} \\ \mathbf{a}_2^T \mathbf{x} \\ \vdots \\ \mathbf{a}_m^T \mathbf{x} \end{bmatrix}.$$

In other words, the i th entry of $\mathbf{y}$ is equal to the inner product of the i th row of A and $\mathbf{x}$, $y_i = \mathbf{a}_i^T \mathbf{x}$.

Matrix Multiplication: Matrix-Vector Products

Given a matrix $A \in \mathbb{R}^{m \times n}$ and a vector $\mathbf{x} \in \mathbb{R}^n$, their product is a vector $\mathbf{y} = A\mathbf{x} \in \mathbb{R}^m$.

- Alternatively, let's write A in column form. In this case we see that,

$$\mathbf{y} = A\mathbf{x} = \begin{bmatrix} | & | & \dots & | \\ \mathbf{a}^1 & \mathbf{a}^2 & \dots & \mathbf{a}^n \\ | & | & & | \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = x_1\mathbf{a}^1 + x_2\mathbf{a}^2 + \dots + x_n\mathbf{a}^n.$$

In other words, $\mathbf{y}$ is a linear combination of the columns of A , where the coefficients of the linear combination are given by the entries of $\mathbf{x}$.

Practice: Two Views of $A\mathbf{x}$

Let

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 5 \\ 6 \end{bmatrix}.$$

- 1 Compute $A\mathbf{x}$ using row–vector inner products.
- 2 Express the same result as a linear combination of the columns of A .

Solution: Two Views of Ax

Using rows,

$$Ax = \begin{bmatrix} 1(5) + 2(6) \\ 3(5) + 4(6) \end{bmatrix} = \begin{bmatrix} 17 \\ 39 \end{bmatrix}.$$

Using columns,

$$Ax = 5 \begin{bmatrix} 1 \\ 3 \end{bmatrix} + 6 \begin{bmatrix} 2 \\ 4 \end{bmatrix} = \begin{bmatrix} 17 \\ 39 \end{bmatrix}.$$

Takeaway

Rows explain individual output entries; columns explain the output as a linear combination.

Matrix Multiplication: Matrix-Matrix Products

Let's look at different ways of viewing the matrix-matrix multiplication
 $C = AB$.

- First, we can view matrix-matrix multiplication as a set of vector-vector products.

$$C = AB = \begin{bmatrix} - & \mathbf{a}_1^T & - \\ - & \mathbf{a}_2^T & - \\ & \vdots & \\ - & \mathbf{a}_m^T & - \end{bmatrix} \begin{bmatrix} | & | & & | \\ \mathbf{b}^1 & \mathbf{b}^2 & \dots & \mathbf{b}^p \\ | & | & & | \end{bmatrix} = \begin{bmatrix} \mathbf{a}_1^T \mathbf{b}^1 & \mathbf{a}_1^T \mathbf{b}^2 & \dots & \mathbf{a}_1^T \mathbf{b}^p \\ \mathbf{a}_2^T \mathbf{b}^1 & \mathbf{a}_2^T \mathbf{b}^2 & \dots & \mathbf{a}_2^T \mathbf{b}^p \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{a}_m^T \mathbf{b}^1 & \mathbf{a}_m^T \mathbf{b}^2 & \dots & \mathbf{a}_m^T \mathbf{b}^p \end{bmatrix}$$

Matrix Multiplication: Matrix-Matrix Products

Let's look at different ways of viewing the matrix-matrix multiplication $C = AB$.

- Alternatively, we can represent A by columns, and B by rows, which leads to a trickier interpretation of AB as a sum of outer products.

$$C = AB = \begin{bmatrix} | & | & & | \\ a^1 & a^2 & \cdots & a^n \\ | & | & & | \end{bmatrix} \begin{bmatrix} - & b_1^T & - \\ - & b_2^T & - \\ & \vdots & \\ - & b_n^T & - \end{bmatrix} = \sum_{i=1}^n a^i b_i^T$$

Matrix Multiplication: Matrix-Matrix Products

In addition to this, it is useful to know a few basic properties of matrix multiplication at a higher level:

- Matrix multiplication is associative: $(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC})$.
- Matrix multiplication is distributive: $\mathbf{A}(\mathbf{B} + \mathbf{C}) = \mathbf{AB} + \mathbf{AC}$.
- Matrix multiplication is, in general, not commutative; that is, it can be the case that $\mathbf{AB} \neq \mathbf{BA}$. (For example, if $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{B} \in \mathbb{R}^{n \times q}$, the matrix product $\mathbf{BA}$ does not even exist if m and q are not equal!)

Try to verify these properties for yourself. *Hint*: Proving matrix properties by reduction to simple scalar properties will come up often.

Proof: Matrix Multiplication Properties

Assume all products below are well-defined. For every entry (i,j) ,

$$\begin{aligned} ((\mathbf{AB})\mathbf{C})_{ij} &= \sum_{\ell} \left(\sum_k a_{ik} b_{k\ell} \right) c_{\ell j} \\ &= \sum_k a_{ik} \left(\sum_{\ell} b_{k\ell} c_{\ell j} \right) = (\mathbf{A}(\mathbf{BC}))_{ij}. \end{aligned}$$

Thus, $(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC})$. Likewise,

$$\begin{aligned} (\mathbf{A}(\mathbf{B} + \mathbf{C}))_{ij} &= \sum_k a_{ik} (b_{kj} + c_{kj}) \\ &= (\mathbf{AB})_{ij} + (\mathbf{AC})_{ij}. \end{aligned}$$

Therefore, $\mathbf{A}(\mathbf{B} + \mathbf{C}) = \mathbf{AB} + \mathbf{AC}$.

A single counterexample disproves commutativity.

Practice: Matrix Multiplication Is Not Commutative

Let

$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}.$$

- 1 Compute $\mathbf{AB}$.
- 2 Compute $\mathbf{BA}$.
- 3 Does $\mathbf{AB} = \mathbf{BA}$?

Solution: Matrix Multiplication Is Not Commutative

$$\mathbf{AB} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}, \quad \mathbf{BA} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}.$$

Therefore,

$$\mathbf{AB} \neq \mathbf{BA}.$$

Takeaway

Even when both products exist and have the same shape, reversing the order generally changes the answer.

Operations and Properties: Identity Matrix

The identity matrix, denoted $\mathbf{I} \in \mathbb{R}^{n \times n}$, has ones on the diagonal and zeros everywhere else:

$$I_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

It leaves every compatible vector or matrix unchanged. For $\mathbf{A} \in \mathbb{R}^{m \times n}$,

$$\mathbf{A}\mathbf{I}_n = \mathbf{A} = \mathbf{I}_m\mathbf{A}.$$

Intuition

$\mathbf{I}$ is the “do nothing” linear transformation, just as 1 is the multiplicative identity for scalars. Its size comes from the surrounding product.

Operations and Properties: Diagonal Matrices

A diagonal matrix is a matrix where all non-diagonal elements are 0. This is typically denoted $\mathbf{D} = \text{diag}(d_1, d_2, \dots, d_n)$, with

$$D_{ij} = \begin{cases} d_i & i = j \\ 0 & i \neq j \end{cases}$$

Clearly, $\mathbf{I} = \text{diag}(1, 1, \dots, 1)$

Intuition

Multiplying by $\mathbf{D}$ scales coordinate i by d_i . A diagonal matrix changes each coordinate independently without mixing coordinates.

Operations and Properties: Transpose

The transpose of a matrix results from "flipping" the rows and columns. Given a matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$, its transpose, written $\mathbf{A}^T \in \mathbb{R}^{n \times m}$, is the $n \times m$ matrix whose entries are given by

$$(\mathbf{A}^T)_{ij} = \mathbf{A}_{ji}.$$

The transpose exchanges row and column roles. It turns a column vector into a row vector and reverses the order of a product.

The following properties of transposes are easily verified:

- $(\mathbf{A}^T)^T = \mathbf{A}$
- $(\mathbf{AB})^T = \mathbf{B}^T \mathbf{A}^T$
- $(\mathbf{A} + \mathbf{B})^T = \mathbf{A}^T + \mathbf{B}^T$

Operations and Properties: Symmetric Matrices

A square matrix $A \in \mathbb{R}^{n \times n}$ is symmetric if $A = A^T$. It is anti-symmetric if $A = -A^T$. It is easy to show that for any matrix $A \in \mathbb{R}^{n \times n}$, the matrix $A + A^T$ is symmetric and the matrix $A - A^T$ is anti-symmetric.

From this it follows that any square matrix $A \in \mathbb{R}^{n \times n}$ can be represented as a sum of a symmetric matrix and an anti-symmetric matrix, since

$$A = \frac{1}{2} (A + A^T) + \frac{1}{2} (A - A^T)$$

Intuition

In a symmetric matrix, the interaction from coordinate i to j matches the interaction from j to i .

Operations and Properties: The Trace

The trace of a square matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ adds the diagonal entries, which describe how much each coordinate contributes back to itself:

$$\text{tr } \mathbf{A} = \sum_{i=1}^n A_{ii}$$

The trace has the following properties:

- $\text{tr}(\mathbf{A}) = \text{tr}(\mathbf{A}^T)$.
- $\text{tr}(\mathbf{A} + \mathbf{B}) = \text{tr}(\mathbf{A}) + \text{tr}(\mathbf{B})$ and $\text{tr}(t\mathbf{A}) = t \text{tr}(\mathbf{A})$.
- $\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$ when the products are defined.
- $\text{tr}(\mathbf{ABC}) = \text{tr}(\mathbf{BCA}) = \text{tr}(\mathbf{CAB})$.

Intuition

$\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$ because both sides sum the same pairwise products $a_{ij}b_{ji}$.

Operations and Properties: Norms

A norm of a vector $\|\mathbf{x}\|$ is informally a measure of the "length" of the vector. For example, we have the commonly-used Euclidean or ℓ_2 norm,

$$\|\mathbf{x}\|_2 = \sqrt{\sum_{i=1}^n x_i^2}.$$

Note that $\|\mathbf{x}\|_2^2 = \mathbf{x}^T \mathbf{x}$

Intuition

Norms turn vectors into lengths, so $\|\mathbf{x} - \mathbf{y}\|_2$ measures the Euclidean distance between two vectors.

Practice: Transpose, Trace, and Norm

Let

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}.$$

- 1 Compute $\|\mathbf{x}\|_2$.
- 2 Compute $\text{tr}(\mathbf{AB})$ and $\text{tr}(\mathbf{BA})$.
- 3 Write $(\mathbf{AB})^T$ without first multiplying $\mathbf{AB}$.

Solution: Transpose, Trace, and Norm

$$\|\mathbf{x}\|_2 = \sqrt{3^2 + 4^2} = 5.$$

Also,

$$\mathbf{AB} = \begin{bmatrix} 4 & 6 \\ -1 & -3 \end{bmatrix}, \quad \mathbf{BA} = \begin{bmatrix} 2 & 4 \\ 1 & -1 \end{bmatrix},$$

so $\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA}) = 1$. Finally,

$$(\mathbf{AB})^T = \mathbf{B}^T \mathbf{A}^T.$$

Takeaway

Transpose reverses product order, while trace permits cyclic reordering.

Operations and Properties: Norms (Optional)

More formally, a norm is any function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ that satisfies 4 properties:

- 1 For all $x \in \mathbb{R}^n$, $f(x) \geq 0$ (non-negativity).
- 2 $f(x) = 0$ if and only if $x = 0$ (definiteness).
- 3 For all $x \in \mathbb{R}^n$, $t \in \mathbb{R}$, $f(tx) = |t|f(x)$ (homogeneity).
- 4 For all $x, y \in \mathbb{R}^n$, $f(x + y) \leq f(x) + f(y)$ (triangle inequality).

Other examples of norms are the ℓ_1 norm,

$$\|\mathbf{x}\|_1 = \sum_{i=1}^n |x_i|$$

and the ℓ_∞ norm,

$$\|\mathbf{x}\|_\infty = \max_i |x_i|.$$

Operations and Properties: Norms (Optional)

In fact, all three norms presented so far are examples of the family of ℓ_p norms, which are parameterized by a real number $p \geq 1$, and defined as

$$\|\mathbf{x}\|_p = \left(\sum_{i=1}^n |x_i|^p \right)^{1/p}.$$

Norms can also be defined for matrices, such as the Frobenius norm,

$$\|\mathbf{A}\|_F = \sqrt{\sum_{i=1}^m \sum_{j=1}^n A_{ij}^2} = \sqrt{\text{tr}(\mathbf{A}^T \mathbf{A})}$$

Many other norms exist, but they are beyond the scope of this review.

Operations and Properties: Linear Independence

A set of vectors $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\} \subset \mathbb{R}^m$ is said to be (linearly) independent if no vector can be represented as a linear combination of the remaining vectors. Conversely, if one vector belonging to the set can be represented as a linear combination of the remaining vectors, then the vectors are said to be (linearly) dependent. That is, if

$$\mathbf{x}_n = \sum_{i=1}^{n-1} \alpha_i \mathbf{x}_i$$

for some scalar values $\alpha_1, \dots, \alpha_{n-1} \in \mathbb{R}$, then we say that the vectors $\mathbf{x}_1, \dots, \mathbf{x}_n$ are linearly dependent; otherwise, the vectors are linearly independent.

Intuition

Independent vectors contain no redundant direction. Removing any one of them reduces what their linear combinations can represent.

Operations and Properties: Rank

The column rank of a matrix $A \in \mathbb{R}^{m \times n}$ is the size of the largest subset of columns of A that constitute a linearly independent set. In the same way, the row rank is the largest number of rows of A that constitute a linearly independent set.

For any matrix $A \in \mathbb{R}^{m \times n}$, it turns out that the column rank of A is equal to the row rank of A , and so both quantities are referred to collectively as the rank of A , denoted as $\text{rank}(A)$.

Intuition

Rank counts the number of independent directions that survive after applying the matrix.

Operations and Properties: Rank

The following are some basic properties of the rank:

- For $A \in \mathbb{R}^{m \times n}$, $\text{rank}(A) \leq \min(m, n)$. If $\text{rank}(A) = \min(m, n)$, then A is said to be full rank.
- For $A \in \mathbb{R}^{m \times n}$, $\text{rank}(A) = \text{rank}(A^T)$
- (Optional) For $A \in \mathbb{R}^{m \times n}$, $B \in \mathbb{R}^{n \times p}$, $\text{rank}(AB) \leq \min(\text{rank}(A), \text{rank}(B))$.
- (Optional) For $A, B \in \mathbb{R}^{m \times n}$, $\text{rank}(A + B) \leq \text{rank}(A) + \text{rank}(B)$

Practice: Rank

Consider the 2×3 matrix

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \end{bmatrix}.$$

- 1 Row reduce $\mathbf{A}$ and find $\text{rank}(\mathbf{A})$.
- 2 Identify the pivot columns in the original matrix.
- 3 Give a basis for $\mathcal{R}(\mathbf{A})$ and determine whether $\mathbf{A}$ has full rank.

Hint

Use the original columns of $\mathbf{A}$, rather than the columns of its row echelon form, when constructing a basis for the range.

Solution: Rank

Subtract twice the first row from the second:

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \end{bmatrix}.$$

The row echelon form has two nonzero rows, so

$$\text{rank}(\mathbf{A}) = 2.$$

The pivots occur in columns 1 and 3. Therefore,

$$\mathcal{R}(\mathbf{A}) = \text{span} \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 7 \end{bmatrix} \right\}.$$

Because $\min(2, 3) = 2$, $\mathbf{A}$ has full row rank.

Operations and Properties: Inverse

The inverse A^{-1} undoes the linear transformation A . It is the unique matrix satisfying

$$A^{-1}A = I = AA^{-1}.$$

Note that not all matrices have inverses. In particular, we say that A is *invertible* or *non singular* if A^{-1} exists and *non invertible* or *singular* otherwise. In order for a square matrix A to have an inverse A^{-1} , then A must be full rank.

As an example of how the inverse is used, consider the linear system of equations, $Ax = b$ where $A \in \mathbb{R}^{n \times n}$, and $x, b \in \mathbb{R}^n$. If A is nonsingular (i.e., invertible), then $x = A^{-1}b$.

Practice: Inverse

Use the same 2×3 matrix and let $\mathbf{B}$ contain its pivot columns:

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 4 \\ 9 \end{bmatrix}.$$

- 1 Can $\mathbf{A}$ have a two-sided inverse? Explain using its shape.
- 2 Find $\mathbf{B}^{-1}$ and solve $\mathbf{B}\mathbf{z} = \mathbf{b}$.
- 3 Use $\mathbf{z}$ to construct one solution of $\mathbf{A}\mathbf{x} = \mathbf{b}$.

Solution: Inverse

The 2×3 matrix A has no two-sided inverse because it is not square.

For the pivot-column matrix, $\det(\mathbf{B}) = 1(7) - 3(2) = 1$, so

$$\mathbf{B}^{-1} = \begin{bmatrix} 7 & -3 \\ -2 & 1 \end{bmatrix}.$$

Therefore,

$$\mathbf{z} = \mathbf{B}^{-1}\mathbf{b} = \begin{bmatrix} 7 & -3 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ 9 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Because $\mathbf{B}$ uses columns 1 and 3 of A , one solution is

$$\mathbf{x} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{Ax} = \begin{bmatrix} 4 \\ 9 \end{bmatrix} = \mathbf{b}.$$

Operations and Properties: Orthogonal Matrices

- Two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ are orthogonal if $\mathbf{x}^T \mathbf{y} = 0$.
- A vector $\mathbf{x} \in \mathbb{R}^n$ is normalized if $\|\mathbf{x}\|_2 = 1$.
- A square matrix $\mathbf{U} \in \mathbb{R}^{n \times n}$ is orthogonal if all its columns are orthogonal to each other and are normalized (the columns are then referred to as being orthonormal).

It follows immediately from the definition of orthogonality and normality that

$$\mathbf{U}^T \mathbf{U} = \mathbf{I} = \mathbf{U} \mathbf{U}^T.$$

Intuition

Square orthogonal matrices represent rotations, reflections, or combinations of both. Their inverse is simply $\mathbf{U}^T$.

Operations and Properties: Orthogonal Matrices

Note that if U is not square - i.e., $U \in \mathbb{R}^{m \times n}$, $n < m$ - but its columns are still orthonormal, then $U^T U = I$, but $U U^T \neq I$.

Another nice property of orthogonal matrices is that operating on a vector with an orthogonal matrix will not change its Euclidean norm, i.e.,

$$\|Ux\|_2 = \|x\|_2$$

for any $x \in \mathbb{R}^n$, $\forall U \in \mathbb{R}^{n \times n}$ orthogonal.

Operations and Properties: Span

The span of a set of vectors $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$ is the set of all vectors that can be expressed as a linear combination of $\{\mathbf{x}_1, \dots, \mathbf{x}_n\}$. That is,

$$\text{span}(\{\mathbf{x}_1, \dots, \mathbf{x}_n\}) = \left\{ \mathbf{v} : \mathbf{v} = \sum_{i=1}^n \alpha_i \mathbf{x}_i, \quad \alpha_i \in \mathbb{R} \right\}$$

It can be shown that if $\{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ is a set of n linearly independent vectors, where each $\mathbf{x}_i \in \mathbb{R}^n$, then $\text{span}(\{\mathbf{x}_1, \dots, \mathbf{x}_n\}) = \mathbb{R}^n$. In other words, any vector $\mathbf{v} \in \mathbb{R}^n$ can be written as a linear combination of $\mathbf{x}_1$ through $\mathbf{x}_n$.

Intuition

The span is the full set of destinations reachable by choosing different coefficients.

Operations and Properties: Range

The range (sometimes also called the columnspace) of a matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$, denoted $\mathcal{R}(\mathbf{A})$, is the span of the columns of $\mathbf{A}$. In other words,

$$\mathcal{R}(\mathbf{A}) = \{\mathbf{v} \in \mathbb{R}^m : \mathbf{v} = \mathbf{A}\mathbf{x}, \mathbf{x} \in \mathbb{R}^n\}.$$

Intuition

$\mathcal{R}(\mathbf{A})$ contains every output that $\mathbf{A}$ can produce.

Operations and Properties: Determinant (Reading)

Intuition: $|A|$ measures signed volume scaling. A zero determinant means that A collapses at least one dimension and is not invertible.

Before giving the general definition for the determinant, we define, for $A \in \mathbb{R}^{n \times n}$, $A_{\setminus i, \setminus j} \in \mathbb{R}^{(n-1) \times (n-1)}$ to be the matrix that results from deleting the i th row and j th column from A . The general (recursive) formula for the determinant is

$$\begin{aligned} |A| &= \sum_{i=1}^n (-1)^{i+j} a_{ij} |A_{\setminus i, \setminus j}| \quad (\text{for any } j \in 1, \dots, n) \\ &= \sum_{j=1}^n (-1)^{i+j} a_{ij} |A_{\setminus i, \setminus j}| \quad (\text{for any } i \in 1, \dots, n) \end{aligned}$$

Operations and Properties: Quadratic Form

Given a square matrix $A \in \mathbb{R}^{n \times n}$ and a vector $x \in \mathbb{R}^n$, the scalar value $x^T A x$ is called a quadratic form. Written explicitly, we see that

$$x^T A x = \sum_{i=1}^n x_i (A x)_i = \sum_{i=1}^n x_i \left(\sum_{j=1}^n A_{ij} x_j \right) = \sum_{i=1}^n \sum_{j=1}^n A_{ij} x_i x_j$$

Note that,

$$x^T A x = (x^T A x)^T = x^T A^T x = x^T \left(\frac{1}{2} A + \frac{1}{2} A^T \right) x,$$

Intuition

$x^T A x$ measures the effect of A along direction x .

Operations and Properties: PSD Matrices

- A symmetric matrix $A \in \mathbb{S}^n$ is positive definite (PD) if for all non-zero vectors $x \in \mathbb{R}^n$, $x^T A x > 0$. This is usually denoted $A \succ 0$.
- A symmetric matrix $A \in \mathbb{S}^n$ is positive semidefinite (PSD) if for all non-zero vectors $x \in \mathbb{R}^n$, $x^T A x \geq 0$. This is usually denoted $A \succeq 0$.

Intuition

A PSD matrix assigns non-negative “energy” to every direction. A PD matrix assigns strictly positive energy to every non-zero direction.

Operations and Properties: PSD Matrices

There is one type of positive semidefinite matrix that comes up frequently, and so deserves some special mention. Given any matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$ (not necessarily symmetric or even square), the matrix $\mathbf{G} = \mathbf{A}^T \mathbf{A}$ (sometimes called a Gram matrix) is always positive semidefinite.

Practice: A Gram Matrix Is PSD

Let

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{G} = \mathbf{A}^T \mathbf{A}.$$

- 1 Compute $\mathbf{G}$.
- 2 Show directly that $\mathbf{x}^T \mathbf{G} \mathbf{x} \geq 0$ for every $\mathbf{x} \in \mathbb{R}^2$.
- 3 Does this argument require computing eigenvalues?

Solution: A Gram Matrix Is PSD

$$\mathbf{G} = \mathbf{A}^T \mathbf{A} = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}.$$

For $\mathbf{x} = \begin{bmatrix} x_1 & x_2 \end{bmatrix}^T$,

$$\mathbf{x}^T \mathbf{G} \mathbf{x} = \mathbf{x}^T \mathbf{A}^T \mathbf{A} \mathbf{x} = \|\mathbf{A} \mathbf{x}\|_2^2 = (x_1 + 2x_2)^2 + x_2^2 \geq 0.$$

Takeaway

Every matrix of the form $\mathbf{A}^T \mathbf{A}$ is PSD; no eigenvalue computation is needed.

Operations and Properties: Eigenvalues & Eigenvectors

Given a square matrix $A \in \mathbb{R}^{n \times n}$, we say that $\lambda \in \mathbb{C}$ is an eigenvalue of A and $\mathbf{x} \in \mathbb{C}^n$ is the corresponding eigenvector if

$$A\mathbf{x} = \lambda\mathbf{x}, \quad \mathbf{x} \neq \mathbf{0}.$$

We can rewrite the equation above to state that $(\lambda, \mathbf{x})$ is an eigenvalue-eigenvector pair of A if,

$$(\lambda I - A)\mathbf{x} = \mathbf{0}, \quad \mathbf{x} \neq \mathbf{0}.$$

Intuition

An eigenvector is a direction that A does not turn. The eigenvalue tells how much A stretches, shrinks, or reverses that direction.

Operations and Properties: Eigenvalues & Eigenvectors

The equation $(\lambda \mathbf{I} - \mathbf{A})\mathbf{x} = 0$ has a non-zero solution if and only if $(\lambda \mathbf{I} - \mathbf{A})$ is singular, i.e.,

$$|(\lambda \mathbf{I} - \mathbf{A})| = 0.$$

We can now use the previous definition of the determinant to expand this expression $|(\lambda \mathbf{I} - \mathbf{A})|$ into a (very large) polynomial in λ , where λ will have degree n . It's often called the characteristic polynomial of the matrix $\mathbf{A}$.

We then find the n (possibly complex) roots of this characteristic polynomial and denote them by $\lambda_1, \dots, \lambda_n$. These are all the eigenvalues of the matrix $\mathbf{A}$, but we note that they may not be distinct.

Operations and Properties: Eigenvalues & Eigenvectors

Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ have eigenvalues $\lambda_1, \dots, \lambda_n$, counted with algebraic multiplicity.

- The trace is the sum of the eigenvalues:

$$\text{tr}(\mathbf{A}) = \sum_{i=1}^n \lambda_i.$$

- The determinant is their product:

$$|\mathbf{A}| = \prod_{i=1}^n \lambda_i.$$

- If $\mathbf{A}$ is real symmetric, $\text{rank}(\mathbf{A})$ equals the number of non-zero eigenvalues.

Operations and Properties: Eigendecomposition(1)

Assume that $A \in \mathbb{R}^{n \times n}$ is a symmetric real matrix. We have the following properties:

- All eigenvalues of A are real numbers. We denote them by $\lambda_1, \dots, \lambda_n$.
- There exists a set of eigenvectors $\mathbf{u}_1, \dots, \mathbf{u}_n$ such that a) for all i , $\mathbf{u}_i$ is an eigenvector with eigenvalue λ_i and b) $\mathbf{u}_1, \dots, \mathbf{u}_n$ are unit vectors and orthogonal to each other.

Let U be the orthonormal matrix that contains $\mathbf{u}_i$'s as columns:

$$U = \begin{bmatrix} | & | & \cdots & | \\ \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_n \\ | & | & \cdots & | \end{bmatrix}$$

Operations and Properties: Eigendecomposition(2)

Let $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$ be the diagonal matrix that contains $\lambda_1, \dots, \lambda_n$ as entries on the diagonal. We can verify that

$$AU = U\Lambda$$

Recalling that orthonormal matrix U satisfies that $UU^T = I$ and using the equation above, we have

$$A = AUU^T = U\Lambda U^T$$

Intuition

U^T moves into the eigenvector coordinate system, Λ scales each coordinate independently, and U moves back.

Practice: Non-negative Eigenvalues and PSD

Assume that $A \in \mathbb{R}^{n \times n}$ is a symmetric real matrix. If the eigenvalues of A are all non-negative, show that A is PSD.

Solution: Non-negative Eigenvalues Imply PSD

Because A is real and symmetric,

$$A = U\Lambda U^T,$$

where U is orthogonal and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$. For any $\mathbf{x}$, let $\mathbf{z} = U^T \mathbf{x}$. Then

$$\mathbf{x}^T A \mathbf{x} = \mathbf{z}^T \Lambda \mathbf{z} = \sum_{i=1}^n \lambda_i z_i^2 \geq 0,$$

because every $\lambda_i \geq 0$. Therefore, $A \succeq 0$.

Practice: Maximizing a Quadratic Form

For a matrix $A \in \mathbb{S}^n$, consider the following maximization problem,

$$\max_{\mathbf{x} \in \mathbb{R}^n} \mathbf{x}^T A \mathbf{x} \quad \text{subject to } \|\mathbf{x}\|_2^2 = 1$$

i.e., find the normalized vector which maximizes the quadratic form.

Solution: Maximizing a Quadratic Form

Write $A = U\Lambda U^T$ and let $z = U^T x$. Since U is orthogonal, $\|z\|_2 = \|x\|_2 = 1$. Thus,

$$x^T A x = \sum_{i=1}^n \lambda_i z_i^2 \leq \lambda_{\max} \sum_{i=1}^n z_i^2 = \lambda_{\max}.$$

Equality holds when z puts all its weight on an index associated with $\lambda_{\max}$. Therefore,

$$\max_{\|x\|_2=1} x^T A x = \lambda_{\max},$$

and a maximizing x is a unit eigenvector associated with $\lambda_{\max}$.

Matrix Calculus: The Gradient

Suppose that $f : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$ is a function that takes as input a matrix $\mathbf{A}$ of size $m \times n$ and returns a real value. Then the gradient of f (with respect to $\mathbf{A} \in \mathbb{R}^{m \times n}$) is the matrix of partial derivatives, defined as:

$$\nabla_{\mathbf{A}} f(\mathbf{A}) \in \mathbb{R}^{m \times n} = \begin{bmatrix} \frac{\partial f(\mathbf{A})}{\partial A_{11}} & \frac{\partial f(\mathbf{A})}{\partial A_{12}} & \cdots & \frac{\partial f(\mathbf{A})}{\partial A_{1n}} \\ \frac{\partial f(\mathbf{A})}{\partial A_{21}} & \frac{\partial f(\mathbf{A})}{\partial A_{22}} & \cdots & \frac{\partial f(\mathbf{A})}{\partial A_{2n}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f(\mathbf{A})}{\partial A_{m1}} & \frac{\partial f(\mathbf{A})}{\partial A_{m2}} & \cdots & \frac{\partial f(\mathbf{A})}{\partial A_{mn}} \end{bmatrix}$$

i.e., an $m \times n$ matrix with

$$(\nabla_{\mathbf{A}} f(\mathbf{A}))_{ij} = \frac{\partial f(\mathbf{A})}{\partial A_{ij}}.$$

Matrix Calculus: The Gradient

Note that the size of $\nabla_A f(\mathbf{A})$ is always the same as the size of $\mathbf{A}$. So if, in particular, $\mathbf{A}$ is just a vector $\mathbf{x} \in \mathbb{R}^n$,

$$\nabla_{\mathbf{x}} f(\mathbf{x}) = \begin{bmatrix} \frac{\partial f(\mathbf{x})}{\partial x_1} \\ \frac{\partial f(\mathbf{x})}{\partial x_2} \\ \vdots \\ \frac{\partial f(\mathbf{x})}{\partial x_n} \end{bmatrix}$$

Matrix Calculus: The Hessian

Suppose that $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a function that takes a vector in $\mathbb{R}^n$ and returns a real number. Then the Hessian matrix with respect to $\mathbf{x}$, written $\nabla_{\mathbf{x}}^2 f(\mathbf{x})$ or simply as H is the $n \times n$ matrix of partial derivatives,

$$\nabla_{\mathbf{x}}^2 f(\mathbf{x}) \in \mathbb{R}^{n \times n} = \begin{bmatrix} \frac{\partial^2 f(\mathbf{x})}{\partial x_1^2} & \frac{\partial^2 f(\mathbf{x})}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f(\mathbf{x})}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f(\mathbf{x})}{\partial x_2 \partial x_1} & \frac{\partial^2 f(\mathbf{x})}{\partial x_2^2} & \cdots & \frac{\partial^2 f(\mathbf{x})}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f(\mathbf{x})}{\partial x_n \partial x_1} & \frac{\partial^2 f(\mathbf{x})}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f(\mathbf{x})}{\partial x_n^2} \end{bmatrix}$$

Matrix Calculus: The Hessian

In other words, $\nabla_{\mathbf{x}}^2 f(\mathbf{x}) \in \mathbb{R}^{n \times n}$, with

$$(\nabla_{\mathbf{x}}^2 f(\mathbf{x}))_{ij} = \frac{\partial^2 f(\mathbf{x})}{\partial x_i \partial x_j}.$$

Note that the Hessian is always symmetric, since

$$\frac{\partial^2 f(\mathbf{x})}{\partial x_i \partial x_j} = \frac{\partial^2 f(\mathbf{x})}{\partial x_j \partial x_i}$$

Matrix Calculus: The Hessian

Note that the gradient of a vector is not defined. Therefore, it is not the case that the Hessian is the gradient of the gradient. However, this is almost true in the following sense: If we look at the i th entry of the gradient $(\nabla_{\mathbf{x}} f(\mathbf{x}))_i = \partial f(\mathbf{x}) / \partial x_i$, and take the gradient with respect to $\mathbf{x}$ we get

$$\nabla_{\mathbf{x}} \frac{\partial f(\mathbf{x})}{\partial x_i} = \begin{bmatrix} \frac{\partial^2 f(\mathbf{x})}{\partial x_i \partial x_1} \\ \frac{\partial^2 f(\mathbf{x})}{\partial x_i \partial x_2} \\ \vdots \\ \frac{\partial^2 f(\mathbf{x})}{\partial x_i \partial x_n} \end{bmatrix}$$

which is the i th column (or row) of the Hessian. Therefore,

$$\nabla_{\mathbf{x}}^2 f(\mathbf{x}) = \begin{bmatrix} \nabla_{\mathbf{x}} (\nabla_{\mathbf{x}} f(\mathbf{x}))_1 & \nabla_{\mathbf{x}} (\nabla_{\mathbf{x}} f(\mathbf{x}))_2 & \cdots & \nabla_{\mathbf{x}} (\nabla_{\mathbf{x}} f(\mathbf{x}))_n \end{bmatrix}.$$

Matrix Calculus: Gradient of Linear Functions

For $\mathbf{x} \in \mathbb{R}^n$, let $f(\mathbf{x}) = \mathbf{b}^T \mathbf{x}$ for some known vector $\mathbf{b} \in \mathbb{R}^n$. Then

$$f(\mathbf{x}) = \sum_{i=1}^n b_i x_i$$

Matrix Calculus: Gradient of Quadratic Functions

The quadratic function $f(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$ for $A \in \mathbb{S}^n$:

$$f(\mathbf{x}) = \sum_{i=1}^n \sum_{j=1}^n A_{ij} x_i x_j$$

Matrix Calculus: Hessian of Linear Functions

For $\mathbf{x} \in \mathbb{R}^n$, let $f(\mathbf{x}) = \mathbf{b}^T \mathbf{x}$ for some known vector $\mathbf{b} \in \mathbb{R}^n$. Then

$$f(\mathbf{x}) = \sum_{i=1}^n b_i x_i$$

Matrix Calculus: Hessian of Quadratic Functions

The quadratic function $f(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$ for $A \in \mathbb{S}^n$:

$$f(\mathbf{x}) = \sum_{i=1}^n \sum_{j=1}^n A_{ij} x_i x_j$$

Matrix Calculus: Least Squares

To recap,

- $\nabla_{\mathbf{x}} \mathbf{b}^T \mathbf{x} = \mathbf{b}$
- $\nabla_{\mathbf{x}} \mathbf{x}^T \mathbf{A} \mathbf{x} = 2\mathbf{A} \mathbf{x}$ (if $\mathbf{A}$ symmetric)
- $\nabla_{\mathbf{x}}^2 \mathbf{x}^T \mathbf{A} \mathbf{x} = 2\mathbf{A}$ (if $\mathbf{A}$ symmetric)

Matrix Calculus: Least Squares

Suppose we are given matrices $A \in \mathbb{R}^{m \times n}$ (for simplicity we assume A is full rank) and a vector $\mathbf{b} \in \mathbb{R}^m$ such that $\mathbf{b} \notin \mathcal{R}(A)$. In this situation we will not be able to find a vector $\mathbf{x} \in \mathbb{R}^n$, such that $A\mathbf{x} = \mathbf{b}$, so instead we want to find a vector $\mathbf{x}$ such that $A\mathbf{x}$ is as close as possible to $\mathbf{b}$, as measured by the square of the Euclidean norm $\|A\mathbf{x} - \mathbf{b}\|_2^2$.

Matrix Calculus: Least Squares

Using the fact that $\|\mathbf{x}\|_2^2 = \mathbf{x}^T \mathbf{x}$, we have

$$\begin{aligned}\|\mathbf{Ax} - \mathbf{b}\|_2^2 &= (\mathbf{Ax} - \mathbf{b})^T (\mathbf{Ax} - \mathbf{b}) \\ &= \mathbf{x}^T \mathbf{A}^T \mathbf{Ax} - 2\mathbf{b}^T \mathbf{Ax} + \mathbf{b}^T \mathbf{b}\end{aligned}$$

Questions

Questions